MILITARY SPECIFICATION

TRUCK, CARGO: 1 1/4 TON, 6X6, M561 AND TRUCK, AMBULANCE: 1 1/4 TON, 6X6, M792

1. SCOPE

1.1 Scope. This specification covers high mobility general and special purpose 1 1/4 ton, compression ignition engine powered, six-wheel drive, articulated two-body military trucks having single tires on all wheels. These vehicles shall be capable of swimming inland water-ways, and be air droppable in Phase I airbourne operations.

1.2 Classification. Vehicles shall be of the following models (see 6.2).

Truck, Cargo: 1 1/4 Ton, 6x6, M561.
Truck, Ambulance: 1 1/4 Ton, 6x6, M792.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-A-548	-	Antifreeze, Ethylene Glycol, Inhibited.
VV-B-680	-	Brake Fluid, Automotive.

Military

MIL-P-514	-	Plates; Identification, Instruction, and Marking Blank.
MIL-T-12459	-	Tire, Pneumatic for Military Ground Vehicles.
MIL-T-46729	-	Tire, Pneumatic: 75% Minimum Natural Rubber Content, For Military Ground Vehicles.
MIL-V-62038	-	Vehicles, Wheeled: Preparation for Shipment and Storage of.
MIL-E-62045	-	Engine, Diesel: Three Cylinder, in Line Type, 103 Horsepower.

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MIL-T-62046 - Transmission Assembly; Cargo Truck, M561.

STANDARDS

Federal Std. No. 595 - Colors.

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-193	-	Painting Procedures, Tactical Vehicles, (Tracked and Wheeled).
MIL-STD-210	-	Climatic Extremes for Military Equipment.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	-	Identification Marking of Combat, and Tactical Transport Vehicles.

PUBLICATIONS

Engineering Parts Lists (EPL)

8736407-1	-	Engineering Parts List for Truck, Cargo: 1 1/4 Ton, 6X6, M561.
8736827	-	Engineering Parts List for Truck, Ambulance: 1 1/4 Ton, 6x6, M792.

Master Lists

8736407-1	-	Master List of Supplementary Quality Assurance Provisions for Truck, Cargo: 1 1/4 Ton, 6X6, M561.
8736827	-	Master List of Quality Assurance Provisions for Truck, Ambulance: 1 1/4 Ton, 6x6, M792.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Preproduction vehicle. When specified (see 6.2), a preproduction vehicle(s) an M561 Truck or M792 Ambulance to be furnished under this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. the vehicle(s) when complete shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles

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proposed to be supplied by the vehicle contractor from production facilities and tooling under the supply contract.

3.2 First production vehicle. One of the first 10 M561 trucks and M792 ambulances respectively produced under the contract shall be submitted to representatives of the rocuring activity for inspection and tests specified herein, to determine conformance to the requirements of this specification. Acceptance of the first production vehicle(s) shall not establish the acceptability of the remaining vehicles under contract in conflict with the acceptance levels specified herein. Vehicle sample(s) submitted by the contractor shall be fully representative of units proposed to be furnished throughout the duration of the contract.

3.2.1 Workmanship and reliability. The workmanship shall be of a quality to assure that the vehicle and components thereof are free from any defects that compromise, limit, or reduce the vehicle or component capabilities herein specified. Vehicles manufactured in accordance with this specification shall be subjected to examinations and tests by the Government to prove vehicle capabilities and performance in accordance with the following:

"(a) The vehicle will have a 90 per cent probability at 90% confidence level of completing 10,000 miles of normal operation with only organizational or operator maintenance and 20,000 miles of normal operation without replacement of any automotive component (see 6.8).

(b) During the 20,000 miles of normal vehicle operation, the total active maintenance downtime less operator maintenance shall be less than 500 manhours and the maintenance requirements shall not exceed the capability of the direct support category of maintenance."

3.2.1.1 Reliability operations. To assure that workmanship and reliability requirements are met during the 20,000 miles of operation, the M561 truck and M792 ambulance with a rated payload shall be capable of operating as follows:

(a) 30 percent on paved roads either concrete or asphalt or any combination of the two.
(b) 36 percent on secondary dirt or gravel roads.
(c) 30 percent on cross-country terrain.
(d) 4 percent on Belgian block.

3.3 Materials. Materials as specified herein and in the referenced drawings, specifications, and standards shall be free from all defects and imperfections that might affect the serviceability and function of the finished product.

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3.3.1 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.4, and necessary in the construction of these vehicles, the contractor shall be responsible for using only those products which are listed (by part or drawing number as QPL items or which have been approved for inclusion on such lists. Manufacturer's inspection recores shall specifically list such component(s) name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.4 Construction. Vehicles, components, subassemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with drawings listed or referred to in Engineering Parts Lists (EPL) 8736407-1 for the M561 and 8736827 for the M792. All parts requiring identification shall be marked in accordance with MIL-STD-130.

3.4.1 Seals. When fording or operating amphibiously in areas with fresh or salt water, the seals shall restrict the entrance of foreign matter into the bearings, which are exposed to contamination during these operations. Water contamination of lubricants shall not be more than 2 percent by volume. All bearing seals shall restrict the leaking of lubricants from the bearings.

3.4.2 Engine. The engine conforming to MIL-E-62045 and installed in accordance with applicable drawings, shall operate under all vehicle operating conditions without malfunction, leaking of coolant, lubricants and fuel.

3.4.2.1 Engine air induction system. The air induction system, as installed, shall prevent entrance of foreign matter during normal vehicle operation.

3.4.2.2 Exhaust system. The exhaust system, as installed, shall be gas-tight and leakproof. The concentration of obnoxious fumes in the crew and personnel compartments of the tractor and carrier shall not exceed .01 percent.

3.4.3 Transmission. The transmission conforming to MIL-T-62046 and associated controls when installed in accordance with the applicable drawings shall operate throughout all speed and gear ranges without malfunction or loss of lubricants.

3.4.4 Steering unit. The steering unit when installed in accordance with applicable drawings shall operate throughout all ranges of steering without malfunction, (wander, side pull) and leakage of lubicants under all vehicle operating conditions.

3.4.5 Fuel system. The fuel system shall maintain a supply of fuel to the engine when vehicle is ascending and descending longitudinal grades of 60 percent in forward and reverse gear, and when traversing side slopes up to 40 percent and with each side of vehicle up slope. The fuel system shall be free from leakage.

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3.4.5.1 Fuel tank, lines and connections. Provisions shall be made to assure the internal cleanliness of the fuel tank, lines, and connections of the fuel system prior to initial fueling of the vehicle.

3.4.6 Throttle linkage. Throttle linkage shall be adjusted to permit full range of engine governed speeds. Accelerator pedal shall bottom out to protect the throttle linkage from damage.

3.4.7 Soft top. Unless otherwise specified (see 6.2), the manufacturer shall furnish the soft top and back panel (without doors) for the M561 tractor and the complete enclosure for the M561 carrier.

3.4.8 Body arrangement and articulation. The tractor and carrier bodies of the M561 and M792 vehicles shall pivot on a transverse axis at the connection of the two bodies such that one body can move up or down with respect the other body (pitch) through an angle of $40^{\circ} \pm 2^{\circ}$. In addition the carrier body shall pivot along a longitudinal axis such that it is free to roll about this axis right and left through an angle of $30^{\circ} \pm 2^{\circ}$, and the center axle assembly can roll about its longitudinal axis $15^{\circ} \pm 2^{\circ}$ to the left and right. These motions can occur simultaneously but to a lesser degree than the individual motions.

3.4.9 Electrical.

3.4.9.1 Alternator voltage. The alternator voltage shall be 28 ± 0.2 volts after installation in the vehicle and connected to the vehicle electrical system in accordance with the applicable drawings. Thereafter, the voltage spread shall not exceed 1 volt under all vehicle operating condition (providing the electrical load does not exceed alternator system capacity). The voltage shall be measured at the point where the ignition lead connects to the vehicle electrical system.

3.4.9.2 Lights and accessories. The internal and external lights and accessories as specified on applicable drawings shall operate as specified through all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions.

3.4.10 Controls. When installed in accordance with applicable drawings, all electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions.

3.4.11 Adjustment mechanisms. When installed and adjusted in accordance with applicable drawings, all adjustments mechanisms shall function properly, and maintain adjustment settings during all conditions of vehicle operations.

3.4.12 Welding repairs and burning operations. Welding repairs and burning operations, other than vehicle attachments, shall be approved by the procuring activity prior to accomplishing repairs.

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3.4.13 Stowed equipment. All On Vehicle Equipment (OVE) items shall be capable of being stowed in the spaces provided, and shall not interfere with operations of vehicle or components.

3.4.14 Special kits. When specified (see 6.2 and 6.6) special kits as applicable shall be furnished.

3.4.15 Vehicle loading. Vehicle weights and loads shall be as specified (see 6.7).

3.4.16 Tires. The tires used on this vehicle shall conform to MIL-T-12459 or MIL-T-46729, as specified by procurement document (see 6.2).

3.4.17 Ambulance (M792). All special equipment provided in the ambulance body shall be easily accessible for use, shall not interfere with loading and unloading of litter patients and shall be securely fastened to withstand all vehicle operating conditions.

3.4.17.1 Heater. With an ambient outside temperature of minus 50oF, the ambulance body heater shall maintain a plus 40oF temperature in the closed ambulance body.

3.4.17.2 Litter holding devices. Holding devices in the M792 Ambulance shall securely hold the litters in place during normal vehicle operations and shall provide easy removal of the litters.

3.5 Performance. The M561 Truck and M792 Ambulance with OVE, and serviced shall meet the applicable performance requirements specified herein after a break-in run of not less than 2 road miles conducted by the contractor. Vehicle, serviced and equipped for prevailing climatic conditions, shall operate as specified herein with and without applicable kits. Performance shall be demonstrated on smooth, relatively level, hard-surfaced roads free of loose material, and cross country terrain as required. The vehicle fully loaded or with a simulated load of equal weight shall operate on longitudinal grades up to 60 percent and side slopes up 40 percent. Note: Side slope requirements shall be established for each Signal Corps electrical equipment shelter assemblage (see 6.6) when installed in the carrier of the M561 Truck and M792 ambulance when operating on these grades and slopes the oil pressure shall be not less than 25 psig, as indicated on the vehicle oil pressure gage.

3.5.1 Condition after run. After completion of the break-in run, the engine shall run smoothly. Prior to vehicle being submitted to the Government for acceptance, the vehicle shall be checked to determine if adjustments are correct and readjusted, if required. There shall be no defects indicative of damaged components or maladjustments, that may cause faulty vehicle operation.

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3.5.2 Environmental. The vehicle shall operate under Intermediate Climatic Conditions in ambient air temperatures ranging from plus 105°F down to minus 25°F without external aids, with special equipment under Hot Dry Climatic Conditions in ambient temperatures up to plus 120°F and under Cold Climatic Conditions in ambient air temperatures down to minus 50°F. The complete vehicle when in storage shall withstand climatic extremes as specified in MIL-STD-210, without deterioration that may cause failure of any component part or of the complete vehicle.

3.5.2.1 Intermediate climatic operation. The vehicle engine shall be capable of being started, and normal operation maintained within two minutes, in ambient air temperatures from minus 25°F to plus 115°F, without external aid, in altitudes from sea level to a 3000 foot elevation above sea level.

3.5.2.1.1 Low temperature operation. When specified (see 6.2), arctic kit shall be furnished and installed. With kit installed and operating, the vehicle engine shall start within 1 minute, and normal operation maintained, in ambient air temperature of minus 50°F and a relative humidity of 100 percent under stand-by operation not to exceed 16 hours duration.

3.5.2.1.2 High temperature operation. The vehicle engine shall start and maintain normal operation in ambient air temperatures and altitudes specified in Table I, without external aids, and with a relative humidity as low as 5 percent.

Table I. Elevation Temperature Chart

Elevation	Minimum ambient air temperature
4,000 feet	110°F
5,000 feet	105.5°F
6,000 feet	101°F
7,000 feet	96.5°F
8,000 feet	92°F

3.5.3 Vapor lock. The vehicle fuel system shall function without evidence of vapor lock, and the engine coolant temperature shall remain below the boiling point with the pressurized radiator cap.

3.5.4 Coolant and lubricant temperatures. The engine coolant (external coolant system pressure per square inch absolute (psia), and oil temperatures, transmission, axles, and transfer case lubricant temperatures, shall not exeeed that specified in Table II, during any of the operational requirements with and without rated payload.

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Table II. Operating Temperatures

Maximum engine coolant temperature measured at coolant outlet with a 31.0 psia ± 2 psia external coolant system pressure	Maximum lubricant temperature measured at engine oil sump	Maximum lubricant temperature measured at transmission sump	Maximum transfer case lubricant temperature measured at transfer sump	Maximum lubricant temperature measured at differential sump
230°F	275°F	300°F	300°F	300°F

3.5.5 Speeds.

3.5.5.1 Level road speeds. The M561 Truck or M792 Ambulance with rated payload, shall maintain a speed of not less than 50 miles per hour (MPH), and maintain a speed of not more than 2-1/2 MPH in low gear without malfunction of any component that may cause vehicle failure.

3.5.5.2 Grade speeds. The M561 Truck or M792 Ambulance fully equipped and with rated payload, but without towed loads shall ascend longitudinal grades up to 60 per cent at a sustained speed of 2 miles per hour. The vehicle shall be stopped on the 60 per cent grade for 1 minute then proceed to ascend the 60 per cent grade.

3.5.5.2.1 With and without towed load. The M561 Truck with rated payload and towed load shall ascend a 10 per cent grade at a sustained speed of 15 MPH. The M561 Truck or M792 Ambulance with rated payload and without towed load shall ascend a 10 per cent grade at a sustained speed of 25 MPH.

3.5.6 Cross-country operation. The M561 Truck shall transport the rated payload and towed load over unimproved roads, trails, open fields, hills and rough cross-country terrain. The M792 Ambulance shall transport only the rated payload over the same terrain.

3.5.7 Braking ability.

3.5.7.1 Service brakes. The service brakes shall stop and hold either fully equipped vehicle with rated payload, traveling 20 MPH within 30 feet from the point of brake application on a surface free of loose material. The service brakes shall control, stop, and hold either vehicle on a hard surface when ascending and descending a grade of 60 per cent. Brake fluid shall conform to VV-B-680.

3.5.7.2 Parking brake. The parking brake shall hold the vehicle with rated payload on a dry concrete grade of 60 percent.

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3.5.8 Cramping angle (turning). Either vehicle shall negotiate full 360 degree turns to the right and left with a turning diameter of 58.0 feet, measured from wall to wall.

3.5.9 Swimming. Either vehicle with and without rated payloads shall swim in water for a period of not less than 20 minutes without special equipment and adjustments or other preparations prior to entering the water.

3.5.9.1 Leakage. Leakage of water into either vehicle shall not exceed 50 pounds or 6 gallons in the tractor, and 10 pounds or 1 1/4 gallons in the cargo compartments of the carrier during a period of 20 minutes. The bilge pump may be used to prolong swimming operations. Either vehicle shall meet the swimming requirements at 2,500 mile intervals during the 20,000 miles of operation. The engine shall be stopped for a period of not less than 1/2 minute during swimming operations and then shall restart immediately.

3.6 Radio suppression. Unless otherwise specified (see 6.2), the M561 Truck and M792 Ambulance shall be radio interference suppressed in accordance with tactical vehicle requirements Class III A of MIL-STD-461.

3.7 Painting, marking and data plates.

3.7.1 Painting. Unless otherwise specified, the exterior and interior of the vehicle, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with MIL-STD-193. Army vehicle exterior paint and exterior nonskid paint shall match color chip X24087 of FEDERAL STD NO. 595.

3.7.2 Marking. Marking shall be in accordance with MIL-STD-642.

3.7.3 Name, shipping, and identification data plates. All plates shall be in accordance with MIL-P-514.

3.8 Servicing and adjustment. The contractor shall service the vehicle for operational use prior to delivery. Such servicing shall include, as a minimum, the focusing of lights, the proper adjustment of the engine, electrical system, brake system; front and rear wheel alignment and balancing of all six wheels; and check for proper tire pressure. Unless otherwise specified, the chassis, engine and all running gear shall be serviced with lubricants of the proper grade for the climatic conditions at the delivery point. Engine cooling system shall be serviced with a solution of ethylene glycol conforming to Type I of O-A-548, and water in equal parts by volume.

3.9 Workmanship. All bolts, nuts or other fasteners shall be torqued as indicated on drawings, or where not specifically detailed on drawings, to the extent consistent with their respective application in commercial vehicles of similar construction.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality control system. Unless otherwise specified by the Procuring Activity the contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the Contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been taken or until conformance of product to specified criteria has been demonstrated.

4.1.3 Materials and qualified products. The contractor shall provide the Government with evidence that materials (see 3.3) used are as specified in the applicable specifications and drawings. The contractor's inspection records shall be checked to determine contractors conformance to 3.3.1. Items produced as specified in QPL will be inspected and tested for conformance to acceptance tests specified in detail specifications less qualification testing.

4.1.4 Parts and components. Parts, components and assemblies shall be inspected in accordance with applicable drawings and Master List of Supplementary Quality Assurance Provisions 8736407-1 or 8736827 as applicable.

4.2 Preproduction and first production inspection.

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4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1) when furnished shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 First production vehicle inspection.

4.2.2.1 Special process inspection. A special process inspection will be conducted by the Government representatives during fabrication of the first production vehicle (see 3.2) to evaluate conformance of materials and workmanship with contract, drawings, and specification requirements.

4.2.2.2 Vehicle inspection. The vehicle or other pertinent components or assemblies shall be inspected at the contractor's or subcontractor's plant prior to covering the weldments (if any) with paint.

4.2.2.2.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be conducted on the following: material, welding, surface hardening, heat treating, protective finishes.

4.2.2.3 Completed vehicle inspection. The first production vehicle (see 3.2) shall be road tested and inspected by the contractor for conformance to requirements of contract and this specification. The road test shall include the two mile break-in (see 3.5) and a five mile (minimum) road test with payload. The vehicle will then be submitted to the Government for inspection.

4.2.2.3.1 Corrections. Corrections found necessary to be made on the vehicle as a result of above inspection shall be made by the contractor.

4.2.2.3.2 Vehicle disposition. Upon completion of inspection and acceptance of the first production vehicles, the vehicles shall remain at the manufacturing facility as a production sample for acceptance criteria. This vehicle may be released for shipment 180 days after start of production has elapse or after successively completion of the 20,000 mile production vehicle test. If the quality level of the vehicles produced, subsequent to first vehicle, exceeds the acceptance standard, the first production vehicle may be released sooner at the discretion of the procuring activity. The contractor shall service and maintain the vehicle during this period in accordance with the applicable documents for care and preservation while in storage.

4.2.3 Production vehicle test. After inspection, correction and acceptance of the first production vehicle, 7 production vehicles as produced, shall be selected by the Government and shall be subjected to examinations and tests for conformance to this specification and Table III - 20,000 mile test. The test will require a minimum of 90 days at a Government test site. The contractor

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shall support the test by providing any vehicle spare parts necessary for the conditions of completion of such a test. Delay in test completion caused by the vehicle breakdown or failure of the contractor to adequately support the vehicle with parts during the tests shall not be the basis for adjustment of the contract delivery schedule or contract price.

Table III. 20,000 Mile Test

Course	Mileage & Speeds	Vehicle Payload
Highway	6,000 miles on paved roads at varying speeds up to 55 mph	Rated payload
Secondary roads	7,200 miles at speeds applicable to conditions of terrain	Rated payload
Cross-country	6,000 miles at speeds applicable to to conditions of terrain	Rated payload
Belgian Block	800 miles at speeds applicable to conditions of terrain	Rated payload
Swimming operation	Every 2,500 miles during this test	Rated payload

NOTE: Except during the swimming operation 50 percent of the test miles shall be run with towed loads as specified in 6.7.

4.2.3.1 Test failure. Failure of test vehicles to comply with the requirements specified or any deficiency of a workmanship or materials nature during or as a result of the 20,000 mile test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 20,000 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the 20,000 mile test are similarly deficient unless evidence satisfactory to the Government is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 INSPECTION PROVISIONS.

4.3.1 Lot size. An inspection lot shall consist of all vehicles of one type, from one day's production, submitted at one time for acceptance inspection.

4.3.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100 percent inspect the first 20 complete vehicles to establish an acceptable process average (see 6.3) to allow normal sampling in accordance with MIL-STD-105.

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4.3.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptance quality levels shown in Table IV and V. The following constitutes a part of the classification of defects (see 6.4):

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any droplet that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

Table IV. Major AQL = 25 Defects/100 Units

Categories		Defects	Method of Inspection
Major:			
101	Steering mechanism	Malfunction; turning radius non-conformance	Visual and functional
102	Power train	Malfunction of clutch, transfer case, transmission, propeller shaft, differential or wheel assembly; lubricant leaks (see 6.4C).	Visual and functional
103	Engine	Malfunction; leaks (see 6.4C); misalinement	Visual and functional
104	Fuel system	Malfunction; damage; leaks (see 6.4a)	Visual and functional
105	Cooling system	Malfunction; damage; leaks (see 6.4b);	Visual
106	Electrical system	Malfunction; damage	Visual and functional
107	Braking ability (service & Parking)	Stopping distance and grade holding nonconformance; malfunction	Visual and functional
108	Speed	Non-conformance	Functional
109	Controls and adjusting mechanisms	Malfunction; improper clearance	Visual and functional

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Table V. Minor AQL = 150 Defects/100 Units

	Categories	Defects	Method of Inspection
Minor:			
201	Steering mechanism	Adjustments improper	Visual and functional
202	Braking ability (service and parking)	Adjustments improper	Visual and functional
203	Controls and adjusting mechanisms	Improper assembly and installation	Visual and functional
204	Fuel system	Improper assembly and installation	Visual and functional
205	Electrical system	Components improperly mounted or adjusted	Visual and functional
206	Decals, marking data, instruction plates	Incomplete; data missing; improper location and size	Visual
207	Paint	Application and color improper	Visual
208	Lubricant	Level and type improper	Visual
209	Ambulance body	Improper installation, malfunction of components	Visual and functional
210	Workmanship	Loose nuts, screws, bolts, washer, cotter pins, safety wires, chain and lock pins, wiring clips, ground straps, grommets, grease fittings; stripped or damaged threads; defective, improperly installed, binding and interference of components. Such defects are considered minor if the location of the fastener(s) does not affect the functioning or the operating life of the vehicle or any of its components.	Visual
211	Protective coatings	Application and coverage improper	Visual
212	Tires	Improper type, inflation improper, size, damage	Visual and Gage

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4.3.2.1 Unclassified defects. All defects having no bearing on function, safety, interchangeability or life, but which are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day period will be added to the minor classification with no increase to AQL's. The deficiency may be deleted from the minor classification of characteristics when five consecutive lots are found free of the deficiency.

4.3.2.2 Recurring major deficiencies. A major deficiency (see 6.5), is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for recurring deficiencies. All deficiencies will be corrected by the contractor.

4.3.2.3 Recurring minor deficiencies. A minor deficiency (see 6.5), is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiency. Correction of all deficiencies shall be accomplished prior to acceptance.

4.4 Classification of tests.

(a) Acceptance tests, 4.4.1 through 4.4.1.16.
(b) Control tests, 4.4.2 through 4.4.2.9.
(c) Comparison tests, 4.4.3 through 4.4.3.9.

4.4.1 Acceptance tests. After the 2 mile break-in (see 3.5) (road), each vehicle selected in accordance with 4.3.1.1, shall be road-tested for a distance of 5 miles or more. Acceptance tests shall include all tests specified at the place of manufacture in Table VI. Vehicles shall be completely assembled, all adjustments completed, and serviced with fuel, lubricants and fluid as specified, but need not be loaded except for braking ability tests as specified in 4.4.2.8. Prior to actual tests, engine shall be operated between 800 and 1200 revolutions per minute to warm engine and lubricants to operating temperature. Operation of vehicle shall be on smooth, level, hard-surfaced roads. Vehicle shall be driven a minimum of 50 feet in reverse gear. After the tests the vehicle shall be examined for evidence of fuel, or lubricant leakage, and other deficiencies.

4.4.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

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Table VI. Location for tests

Title	Paragraph Numbers	Place of Manufacture	Gov't proving ground
Acceptance tests	4.4.1	X	X
Seals check	4.4.1.2	X	X
Engine and components check	4.4.1.3	X	X
Transmission, steering and components check	4.4.1.4	X	X
Fuel system and components check	4.4.1.5	X	X
Throttle linkage check	4.4.1.6	X	X
Alternator voltage, lights and accessories test	4.4.1.7	X	X
Controls and adjustment mechanism check	4.4.1.8	X	X
Welding repairs and burning operations check	4.4.1.9	X	X
Tire check	4.4.1.10	X	X
Ambulance (792) equipment check	4.4.1.11	X	X
Vapor lock check	4.4.1.12	X	X
Level road speed test	4.4.1.13	X	X
Braking ability check	4.4.1.14	X	X
Turning (cramping) checks	4.4.1.15	X	X
Service and adjustment check	4.4.1.16	X	X
Control tests	4.4.2	X	X
Engine and fuel system grade and slope check	4.4.2.1	X	X
Soft top check	4.4.2.2	X	X
Body arrangement and articulation check	4.4.2.3	X	
Stowed equipment check	4.4.2.4	X	X
Special kits check	4.4.2.5	X	X
Vehicle loading, level and grade speed tests	4.4.2.6	X	X
Cross-country test	4.4.2.7	X	X
Braking ability tests	4.4.2.8	X	X
Swimming operational test	4.4.2.9	x	X
Comparison tests	4.4.3		X
Air induction test	4.4.3.2		
Obnoxious fumes concentration check	4.4.3.3		X
Vehicle grade and slope operational test	4.4.3.4		X
Vehicle articulation test	4.4.3.5		X
Environmental operational tests	4.4.3.6		X
Vehicle operational tests	4.4.3.7		X
Swimming operational tests	4.4.3.8		X
Radio interference suppression test	4.4.3.9		X
Preservation, packaging and vehicle processing	4.5	X	

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4.4.1.2 Seals (contamination) check. To determine conformance to 3.4.1, vehicle shall be subjected to fording operation. After fording, all bearing subject to water contamination shall be checked. Contamination not to be more than specified.

4.4.1.3 Engine and components check. To determine conformance to 3.4.2 and 3.4.2.1, during the period of vehicle operation, the engine and components shall be observed for functional requirements.

4.4.1.4 Transmission, steering and components check. To determine conformance to 3.4.3 and 3.4.4, during the period of vehicle operation the transmission and steering components shall be observed for functional requirements.

4.4.1.5 Fuel system cleanliness and component check. To determine conformance to 3.4.5 and 3.4.5.1, the fuel system and components shall be checked for functional requirements and leaks, less grade and slopes operation. Prior to operations the fuel system shall be checked for cleanliness.

4.4.1.6 Throttle linkage check. To determine conformance to 3.4.6, during the period of vehicle operation the throttle linkage shall be observed for functional operation, and after the period the linkage shall be checked for proper adjustment.

4.4.1.7 Alternator voltage, lights, and accessories test. To determine conformance to 3.4.9.1 and 3.4.9.2, the alternator voltage shall be checked at point specified and the voltage shall be as specified. The lights and accessories shall be operated and checked for functional requirements.

4.4.1.8 Controls and adjustment mechanisms check. To determine conformance to 3.4.10 and 3.4.11, the controls shall be operated as required and checked for functional requirements. Adjustment mechanisms shall be checked for proper adjustment.

4.4.1.9 Welding repairs and burning operations check. To determine conformance to 3.4.12, all welding reparis and burning operations shall be examined for conformance to approved welding procedure.

4.4.1.10 Tire check. To determine conformance to 3.4.16, tires shall conform to requirements specified.

4.4.1.11 Ambulance (M792) equipment check. To determine conformance to 3.4.17 through 3.4.17.2, all special equipment installed and the ambulance body inclosure shall be checked for proper installation and function as specified on applicable drawings.

4.4.1.12 Vapor lock check. To determine conformance to 3.5.3, the fuel

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system shall be observed for vapor lock when operating vehicle, and special attention shall be taken, for signs of vapor lock during extreme hot weather conditions.

4.4.1.13 Level road speed test. To determine conformance to 3.5.5.1 the vehicle less payload shall be operated as specified and observed for speed and performance requirements.

4.4.1.14 Braking ability check. To determine conformance to 3.5.7.1 and 3.5.7.2, the ability of the brakes shall be checked for stopping and holding.

4.4.1.15 Cramping angle. To determine conformance to 3.5.8, the vehicle shall be turned to the right and to the left in specified radius. The axle stops shall be checked for proper adjustment.

4.4.1.16 Servicing and adjustment check. To determine conformance to 3.8, the vehicle shall be checked for conformance to requirements specified therein.

4.4.2 Control tests. Control tests for maintaining control of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein. The Government shall select at random for control test 1 vehicle out of each 100 vehicles submitted, but not more than two vehicles in any given 30 day period. Each vehicle selected shall be operated, with designated or simulated payload, for a distance of 50 miles by the contractor. These tests shall be performed in the presence of a Government representative.

4.4.2.1 Engine and fuel system grade and slope check. To determine conformance to 3.4.5 and 3.5, the vehicle shall be placed on grade or slope or to be so positioned as to simulate the grade and slope specified. The engine shall be operated at 800 to 1200 RPM for 5 minutes then stopped for not less than 1 minute, the engine shall restart in less than 1 minute. Oil pressure shall not be less than that specified.

4.4.2.2 Soft top check. To determine conformance to 3.4.7, when a soft top is furnished and installed in accordance with applicable drawings, it shall be checked for completeness and proper installation.

4.4.2.3 Body arrangement and articulation check. To determine conformance to 3.4.8, the vehicle with rated payload shall be operated as specified and observed for performance requirements. Test shall be accomplished using test course as shown on figure 1.

4.4.2.4 Stowed equipment check. To determine conformance to 3.4.13, all manufacture and depot installed "On Vehicle Equipment" (OVE) shall be stowed (combat loaded) on the selected vehicle. The equipment shall be removed after the test. The OVE shall be of the latest configuration available to the contractor, and the contractor shall be permitted to unpackage and repackage for domestic shipment only equipment used for test.

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4.4.2.5 Special kits check. To determine conformance to 3.4.14, kits when furnished shall be checked for completeness and damage.

4.4.2.6 Vehicle loading and level road speed test. To determine conformance to 3.4.15 and 3.5.5.1, the vehicle shall be loaded with applicable rated payload and operated as specified and vehicle observed for performance and speed requirements.

4.4.2.7 Cross-country test. To determine conformance to 3.5.6, the vehicle shall operate as specified without malfunction.

4.4.2.8 Braking ability tests. To determine conformance to 3.5.7.1 and 3.5.7.2, during the 50 mile road test, the ability of the braking systems to stop and hold vehicle shall be as specified.

4.4.2.9 Operational swimming test. To determine conformance to 3.4.1, 3.5.9 and 3.5.9.1, the vehicle shall be operated as specified. Vehicle shall be checked for performance and leaks. Bilge pump shall be operated during swimming test to determine performance and rate of water removal.

4.4.3 Comparison tests. The Government may select vehicles any time during the contract production period and subject vehicle to all tests listed in Table VI and Table VII to determine if any deficiencies of a material or workmanship exist. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a spot check basis. Vehicle selected shall not include any vehicles previously tested for conformance to 4.4.2 (control tests).

4.4.3.1 Test failure. Failure of any vehicle tested to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions.

Table VII. Comparison test (5000 miles)

Course	Mileage and speeds	Vehicle Payload
Highway	100 miles break-in at speeds up to 30 mph. 1,400 miles at variable speed up to 55 mph.	Rated payload
Secondary road	1,800 miles at speeds applicable to conditions of terrain.	Rated payload
Cross-country	1,500 miles at speeds applicable to to conditions of terrain.	Rated payload
Belgian block	200 miles at speeds applicable to conditions of terrain	Rated payload
Swimming operation	Every 2,500 miles during this test	Rated payload

NOTE: Except during the swimming operation 50 percent of the test miles shall be run with towed loads as specified in 6.7.

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4.4.3.2 Air induction system test. To determine conformance to 3.4.2.1, vehicle shall be operated under such conditions that when the air induction system and air cleaners are checked, it can be determined if air induction system has met its functional requirements.

4.4.3.3 Obnoxious fumes concentration check. To determine conformance to 3.4.2.2, the crew compartment shall be checked for concentration of obnoxious fumes. Fumes shall not exceed percent specified.

4.4.3.4 Vehicle grade and slope operational test. To determine conformance to 3.4.5, and 3.4.5.1 and 3.5, the vehicle shall be operated on specified grades and side slopes and checked for performance requirements.

4.4.3.5 Vehicle articulation test. To determine conformance to 3.4.8, the vehicle (tractor and carrier) shall pivot and roll as specified without damage to components, malfunction of engine, transmission and drive train (see figure 1).

4.4.3.6 Environmental operational tests. To determine conformance to 3.5.2 through 3.5.2.1.2, the vehicle properly serviced and equipped shall be subjected to applicable tests in low and high temperatures as required to assure that vehicle meets all the environmental operational requirements specified.

4.4.3.7 Vehicle operational tests. To determine conformance to 3.5.4 through 3.5.6, vehicle shall be operated as specified and checked for speed, cross-country, grade speeds and highway performance, and coolant and lubricant temperatures shown in Table II.

4.4.3.8 Swimming operational tests. To determine conformance to 3.4.1, 3.5.9 and 3.5.9.1, the vehicle shall be operated as specified. Performance and leakage shall be checked. Bilge pump shall be operated during swimming test to determine performance and rate of water removal.

4.4.3.9 Radio interference suppression test. To determine conformance to 3.6, the vehicle shall be subjected to a radio interference suppression test as specified.

4.5 Preservation, packaging and vehicle processing. Material and equipment shall be inspected prior to shipment to assure conformance to the applicable requirements specified in 5.1.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicle and equipment shall be processed for shipment and storage in accordance with MIL-V-62038 to the extent indicated on the applicable vehicle preservation data sheet or other implementation document, as specified by the procuring activity (see 6.2).

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6 NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general use by the United States Services in transporting personnel or cargo, or in providing for special tasks, during tactical military operations. The vehicles are intended for use under extreme or unusual conditions of climate, weather, terrain, and military service which includes air transportation and Phase I tactical air drop by parachute.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Model required (see 1.2).
(c) Preproduction model, required (see 3.1).
(d) Soft top, not required (see 3.4.7).
(e) Special kits, if required (see 3.4.14).
(f) Type of tires required (see 3.4.16).
(g) Arctic kit if required (see 3.5.2.1.1).
(h) Radio suppression, if other than specified (see 3.6).
(i) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process Average} = \frac{\text{Number of Defects}}{\text{Number of Vehicles Inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100% inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.4 The following definitions shall be used with the classification of defect:

(a) Weep: Any evidence of fluid beyond the seal
(b) Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
(d) Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

6.5 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any

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other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the Vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

6.6 Kits. The M561 Truck and M792 Ambulance are designed for compatibility with the following kits (except as noted) when installed in accordance with applicable drawings:

	Nomenclature	Drawing Number
*(1)	Electrical Equipment Shelter S-250	N/A
*(2)	Radio Installations	N/A
(3)	Front Mounted Winch Kit	11595248
(4)	Arctic Kit, -50°F	11595252
(5)	Winterization Kit, -25°F	11595251
(6)	100 Amp Power Source	11594549
*(7)	7.62mm Machine Gun Mount Kit	11595644
(8)	Slave Receptacle Kit	11595525
*(9)	Stowage Kit, Mortar 81mm	11595606
*(10)	Stowage Kit, Mortar 7.62 inch	11595578

*For M561 Truck only.

6.7 Vehicle loading. The following vehicle weights and loads are approximate:

Model Vehicle	Curb Weight	Rated Payload Highway or Cross Country	Gross Vehicle Weight	Rated Towed Load Highway or Cross Country
M561	7,000 lbs	2,900 lbs	10,000	50% of GVW required
M792	7,350 lbs	2,900 lbs	10,250	

6.7.1 Curb weight. Curb weight includes weight of vehicle with all attachments, accessories, OVE and equipment, full complement of fuel, lubricants and coolant.

6.7.2 Payload. Vehicle payload shall include driver, co-driver and 1 1/4 tons of cargo.

6.7.3 Towed load. With towed load the vehicle performance as specified herein shall be met when the M561 is coupled to the M104A1 cargo type trailer. Total weight of trailer and cargo shall be limited to the rated towed loads specified herein.

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6.8 Automotive Components. Critical components on the M561 and M792 are as follows:

SYSTEM	MAJOR AUTOMOTIVE COMPONENTS M561/M792
a. Power Train	Engine Transmission Transfer Assembly Clutch Assembly Differentials Propeller Shafts (Less U-Joints) Drive Axle Shafts (Less U-Joints)
b. Brake System	Master Cylinder Sealed Brake Assemblies
c. Steering System	Gear Boxes & Linkages

Custodian

Army (AT)

Preparing activity

Army (AT)

Project No. 2320-A638

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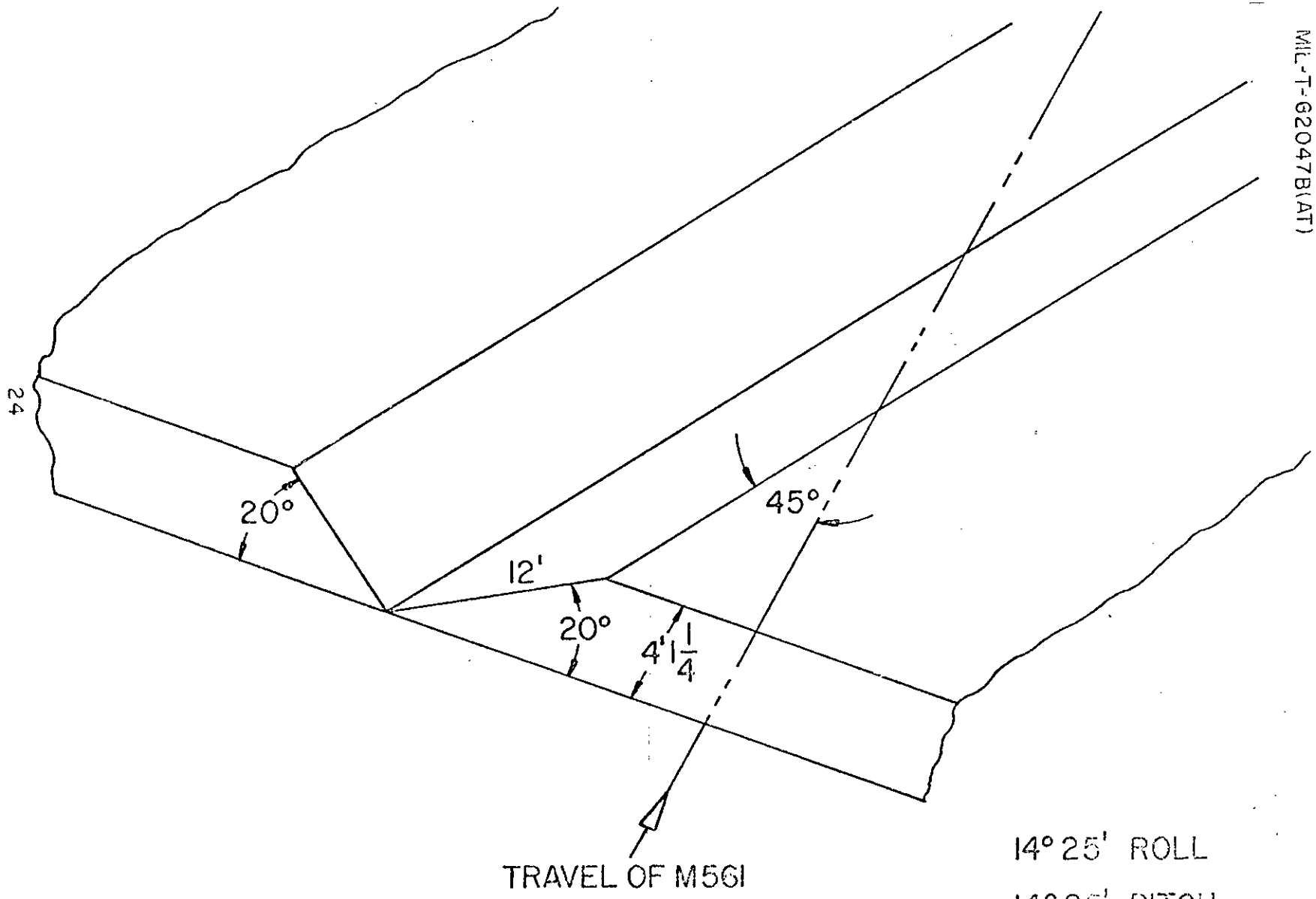


FIGURE I. ARTICULATION TEST COURSE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-62047B(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRARH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to prepering activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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